CIS 106 Final Project — Deliverable 1: Project Requirements

1. What is a web server?

A web server is software that stores, processes, and delivers website content to users through the internet or a local network. Apache is an example of web server software.

2. What are the different web server applications?

- **Apache HTTP Server:** Open-source web server software. It supports Linux, Windows, and other operating systems.
- **NGINX:** Open-source web server software also used as a reverse proxy and load balancer.
- **Microsoft IIS:** Web server software developed by Microsoft. It is included with supported Windows versions and is designed primarily for Windows.
- **Caddy:** Open-source web server known for automatically managing HTTPS certificates.

3. What is virtualization?

Virtualization is technology that allows one physical computer to run multiple virtual computers using software.

4. What is VirtualBox?

VirtualBox is free, open-source virtualization software that allows users to create and run virtual machines on a physical computer.

5. What is a virtual machine?

A virtual machine (VM) is a software-based computer that runs its own operating system and applications using virtual hardware.

6. What are a host machine and a guest machine?

The host machine is the physical computer running virtualization software. The guest machine is the virtual computer running inside that software.

7. What is Debian?

Debian is a free, open-source Linux-based operating system known for its stability, security, and large collection of software packages.

8. What is a firewall?

A firewall is a security system that monitors and controls network traffic based on rules to help prevent unauthorized access.

9. What is SSH?

SSH stands for Secure Shell. It is a network protocol that allows users to securely access and control another computer remotely through an encrypted connection.

10. What is an IP address?

An IP address is a numerical address assigned to a device on a network so it can communicate with other devices.

11. What is a network mask?

A network mask, also called a subnet mask, identifies which part of an IP address represents the network and which part identifies the device.

12. What is a port?

A port is a numbered communication endpoint used by network applications to send and receive data. HTTP commonly uses port 80, HTTPS uses port 443, and SSH uses port 22.

13. What is port forwarding?

Port forwarding directs network traffic arriving at a particular port to a specific device or service. In VirtualBox, it can allow a computer to access a web server running inside a virtual machine.

14. What is localhost?

Localhost is a hostname that refers to the computer currently being used. It is commonly used to access services running on the same machine.

15. What does 127.0.0.1 represent?

127.0.0.1 is the IPv4 loopback address. It allows a computer to communicate with itself without sending traffic to another device.

16. What is Git?

Git is a version control system that tracks changes to files and helps users manage different versions of a project.

17. What is GitHub?

GitHub is an online platform for hosting Git repositories. It allows users to store code and documents, track changes, collaborate with others, and share projects.

Concepts that need more understanding

Virtual Machine Networking: I still have trouble understanding the difference between NAT and Bridged Adapter and when to use each one.

Port Forwarding: I understand that port forwarding allows connections to reach a virtual machine, but I'm still confused about how the ports connect between the host and the server.

Apache Virtual Hosts: I understand that virtual hosts allow one Apache server to host multiple websites, but I'm still learning how to configure each website separately.